

Linear Regression & Data Preparation Quiz

1. What is the fundamental goal of Simple Linear Regression?

- A) To find a curved line that connects all data points.
- B) To find the best-fitting straight line that minimizes the distance (error) between the line and data points.
- C) To split the data into multiple clusters.
- D) To predict a categorical label (e.g., Yes/No).

2. Which Python library is primarily used for creating and manipulating DataFrames?

- A) numpy
- B) matplotlib
- C) pandas
- D) sklearn

3. Before modeling, which method is best used to check if there is a strong linear relationship between variables?

- A) `df.describe()`
- B) `df.info()`
- C) `df.corr()`
- D) `df.head()`

4. In the context of Machine Learning, what does the variable 'X' usually represent?

- A) The Target (Dependent Variable)
- B) The Features (Independent Variables)
- C) The Error rate
- D) The Prediction result

5. What does the variable 'y' usually represent?

- A) The Features (Input)
- B) The Target (What we want to predict)
- C) The Index of the DataFrame
- D) The Random State

6. In scikit-learn, what is the required dimensionality (shape) for the Feature matrix (X)?

- A) 1D array (Vector)
- B) 2D array (Matrix)
- C) 3D array (Tensor)
- D) It does not matter.

7. How should you select a single column "Area" from a DataFrame df to be used as X (Features)?

- A) df['Area']
- B) df.Area
- C) df[['Area']]
- D) df('Area')

8. What is the typical dimensionality of the Target variable (y) in scikit-learn?

- A) 1D array (Series)
- B) 2D array (Matrix)
- C) A Dictionary
- D) A list of lists

9. Why do we perform a Train-Test Split?

- A) To make the dataset smaller for faster processing.
- B) To remove outliers from the data.
- C) To evaluate the model's performance on unseen data and avoid memorization.
- D) To convert categorical data into numbers.

10. Which function is used to split the dataset?

- A) `split_data()`
- B) `train_test_split()`
- C) `separate_train_test()`
- D) `model_selection_split()`

11. What is the correct order of the return values for `train_test_split(X, y)`?

- A) `X_train, y_train, X_test, y_test`
- B) `X_train, X_test, y_train, y_test`
- C) `y_train, y_test, X_train, X_test`
- D) `X_test, X_train, y_test, y_train`

12. What is the purpose of the `random_state` parameter?

- A) To increase the accuracy of the model randomly.
- B) To ensure the data split is reproducible (the same split every time code runs)
- C) To randomize the index of the dataframe.
- D) It determines the percentage of the test set.

13. How do you instantiate (create an object of) a Linear Regression model?

- A) `model = LinearRegression.fit()`
- B) `model = new LinearRegression()`
- C) `model = LinearRegression()`
- D) `model = train_model(LinearRegression(`

14. Which method is used to train the model?

- A) `.train()`
- B) `.predict()`
- C) `.score()`
- D) `.fit()`

15. Which data should be passed to the `.fit()` method?

- A) `X_test, y_test`
- B) `X_train, y_test`
- C) `X_train, y_train`
- D) `X, y` (The entire dataset)

16. What is "Data Leakage" in the context of model training?

- A) When the model is trained on the Test set.
- B) When the dataset has missing values.
- C) When the computer runs out of memory.
- D) When the correlation is negative.

17. After fitting the model, which attribute stores the "slope" (coefficient) of the line?

- A) .slope_
- B) .coef_
- C) .m_
- D) .gradient_

18. After fitting the model, which attribute stores the "intercept" (bias)?

- A) .bias_
- B) .start_
- C) .intercept_
- D) .c_

19. Which method is used to generate predictions for new data?

- A) .guess()
- B) .calc()
- C) .fit()
- D) .predict()

20. Which dataset is typically passed to the .predict() method for evaluation?

- A) y_test
- B) X_train
- C) X_test
- D) y_train